

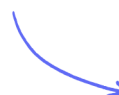
IGCSE • Cambridge (CIE) • Maths

 25 mins  6 questions

Investigation: Hexagonal Numbers

Investigations

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Total Marks

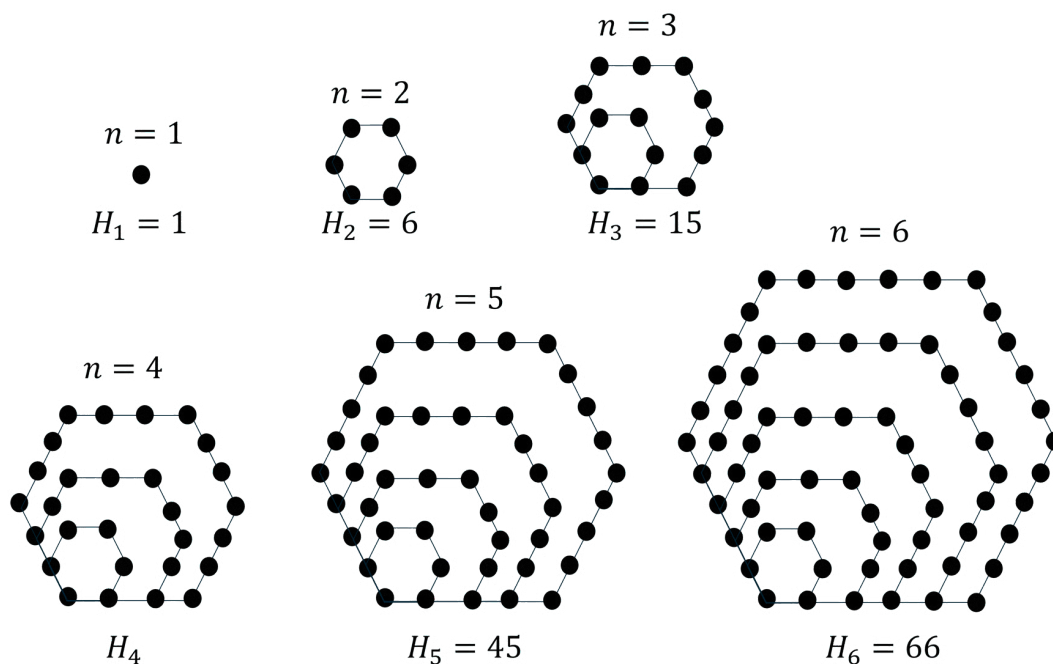
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- 1 This investigation introduces and explores the properties and patterns of the sequence of hexagonal numbers.

Hexagonal numbers are numbers formed by summing equally spaced dots around hexagons that sit inside of each other, as shown below.

If n is the position of the term in the sequence then H_n is the n th hexagonal number.

Hexagonal numbers



By counting the total number of dots in the diagram for $n = 4$, work out the fourth hexagonal number, H_4 .

(1 mark)

- 2 (a) The hexagonal numbers are an increasing sequence of numbers, with each number greater than the one before.

The first six hexagonal numbers are

$$H_1 = 1, H_2 = 6, H_3 = 15, H_4 = 28, H_5 = 45, H_6 = 66$$

Show that the first three hexagonal numbers are triangular numbers.

(2 marks)

(b) Is it true that all triangular numbers are hexagonal numbers?

Explain your answer.

(1 mark)

- 3 (a)** The hexagonal numbers are an increasing sequence of numbers, with each number greater than the one before.

The first six hexagonal numbers are

$$H_1 = 1, H_2 = 6, H_3 = 15, H_4 = 28, H_5 = 45, H_6 = 66$$

To test whether a number, x , is a hexagonal number or not, substitute it into the formula:

$$\frac{1 + \sqrt{8x + 1}}{4}$$

If the output is a positive integer, then it is a hexagonal number.

For example, to test if 6 is a hexagonal number, substitute in $x = 6$:

$$\begin{aligned}\frac{1 + \sqrt{8 \times 6 + 1}}{4} &= \frac{1 + \sqrt{48 + 1}}{4} \\ &= \frac{1 + \sqrt{49}}{4} \\ &= \frac{1 + 7}{4} \\ &= \frac{8}{4} \\ &= 2\end{aligned}$$

The output is 2, which is a positive integer, so 6 is a hexagonal number.

Use the test above to determine whether 120 is a hexagonal number.

You must show your working clearly.

(2 marks)

- (b)** Is it enough for the square root part of the formula, $\sqrt{8x + 1}$, to be an integer in order for the output of the formula to be an integer?

Explain your answer.

(2 marks)

- (c) A student believes that the number 1 is the only square number that is also a hexagonal number.

Use the number 1225 to show that the student is **not** correct.

You must show your working clearly.

(3 marks)

- 4 The hexagonal numbers are an increasing sequence of numbers, with each number greater than the one before.

To test whether a number, x , is a hexagonal number or not, you can substitute it into the formula:

$$\frac{1 + \sqrt{8x + 1}}{4}$$

If the output is a positive integer, then it is a hexagonal number.

A hexagonal number, H_n , and its position value, n , satisfy this formula in the following way:

$$\frac{1 + \sqrt{8H_n + 1}}{4} = n$$

By making H_n the subject, show that the formula for the n th term of a hexagonal number is given by

$$H_n = 2n^2 - n$$

(4 marks)

5 (a) The sequence of hexagonal numbers, H_n , is shown below.

n	1	2	3	4	5	6
H_n	1	6	15	28	45	66

A new sequence, K_n , is formed using the n th term formula:

$$K_n = n + H_n$$

The first three terms, K_1 , K_2 and K_3 , are shown in the table below.

n	1	2	3	4	5	6
H_n	1	6	15	28	45	66
K_n	2	8	18			

Find K_4 , K_5 and K_6 .

(2 marks)

- (b) A fraction is said to be a perfect square if it can be written in the form $\left(\frac{a}{b}\right)^2$ where a and b are non-zero integers.

Terms in the sequence of K_n can be made into fractions by dividing the previous term, K_{n-1} , by the current term:

$$\frac{K_{n-1}}{K_n} = \frac{\text{previous term}}{\text{current term}}$$

Use the table below to show that fractions formed in this way are perfect squares.

The first example has been done for you, as follows:

$$\frac{K_1}{K_2} = \frac{2}{8} = \frac{1}{4} = \left(\frac{1}{2}\right)^2$$

n	1	2	3	4	5	6
H_n	1	6	15	28	45	66
K_n	2	8	18			
$\frac{K_{n-1}}{K_n}$	No previous term	$\left(\frac{1}{2}\right)^2$				

(3 marks)

- (c) By substituting the n th term formula

$$H_n = 2n^2 - n$$

into the formula

$$K_n = n + H_n$$

and simplifying, use algebra to prove that

$$\frac{K_{n-1}}{K_n} = \frac{(n-1)^2}{n^2}$$

(3 marks)

- 6** The hexagonal numbers are an increasing sequence of numbers, with each number greater than the one before.

The n th term formula for the hexagonal numbers is

$$H_n = 2n^2 - n$$

A new sequence is given by

$$R_n = \frac{2}{2n-1} - \frac{1}{n}$$

Use algebra to prove that R_n is the sequence of the reciprocals of hexagonal numbers.

(2 marks)